FSF3827 Assignment 3

Anh Do
020416-2317
anhd@kth.se

December 4, 2025

Question 1

Part 1

Let C be a minimal cover for the knapsack set K. The corresponding valid inequality (cover cut) is:

$$\sum_{i\in C} x_i \leq |C| - 1$$

Let C' be an extension of set C by a single variable, defined as $C' = C \cup \{j\}$ for some $j \notin C$. The cover cut for C' is given by:

$$\begin{aligned}
\sum_{i\in C'} x_i &= \sum_{i\in C} x_i + x_j \leq |C'| - 1 \\
&= \sum_{i\in C} x_i + x_j \leq |C| \\
\implies \quad \sum_{i\in C} x_i &\leq |C| - x_j
\end{aligned}$$

Since $x_j \in \{0,1\}$, we observe that:

$$\sum_{i\in C} x_i \leq |C| - 1 \leq |C| - x_j$$

Thus, the cover cut for C dominates the cover cut for C' because it imposes a more restrictive bound. Consequently, the cover cut for C' is weaker.

Part 2

Lets lift a variable $x_k \notin C'$ into the cover cut for C. The lifted inequality is given by:

$$\sum_{i\in C} x_i + \alpha_{k_1} x_k \leq |C| - 1$$

Where the lifting coefficient α_{k_1} is defined as:

$$\alpha_{k_1} = |C| - 1 - \max\left(\sum_{i\in C} x_i | x_k = 1, x \in \{0,1\}^n, \sum_{i\in N} a_i x_i \leq b, N = C \cup k\right)$$

Similarly, we lift the same variable x_k into the cover cut for C'. The lifted inequality can thus be expressed as:

$$\sum_{i\in C'} x_i + \alpha_{k_2} x_k \leq |C'| - 1 \implies \sum_{i\in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j$$

Here the lifting coefficient α_{k_2} is defined as:

$$\begin{aligned}
\alpha_{k_2} &= |C'| - 1 - \max\left(\sum_{i\in C'} x_i | x_k = 1, x \in \{0,1\}^n, \sum_{i\in N} a_i x_i \le b, N = C' \cup k\right) \\
&= |C| - \max\left(\sum_{i\in C} x_i + x_j | x_k = 1, x \in \{0,1\}^n, \sum_{i\in N} a_i x_i \le b, N = C \cup j \cup k\right) \\
&\ge |C| - \max x_j - \max\left(\sum_{i\in C} x_i | x_k = 1, x \in \{0,1\}^n, \sum_{i\in N} a_i x_i \le b, N = C \cup k\right) \\
&\ge |C| - 1 - \max\left(\sum_{i\in C} x_i | x_k = 1, x \in \{0,1\}^n, \sum_{i\in N} a_i x_i \le b, N = C \cup k\right) \\
&\ge \alpha_{k_1}
\end{aligned}$$

Thus we can see that:

$$\begin{aligned}
&\sum_{i\in C} x_i + \alpha_{k_2} x_k \le |C| - x_j \\
\implies \sum_{i\in C} x_i + \alpha_{k_1} x_k \le &\sum_{i\in C} x_i + \alpha_{k_2} x_k \le |C| - x_j \\
\implies \sum_{i\in C} x_i + \alpha_{k_1} x_k \le &\sum_{i\in C} x_i + \alpha_{k_2} x_k \le |C| - 1 \le |C| - x_j
\end{aligned}$$

Therefore, the lifted cover cut for C dominates the lifted cover cut for C', making the latter weaker.

Question 2

Part 1

Inital LP relaxation solution:

Table 1: Optimization Results (Objective Value: 194.2)

Index (i)	**Variable 1** ($j = 1$)	**Variable 2** ($j = 2$)
1	0.0	1.0
2	0.0	0.0
3	1.0	0.0
4	0.0	0.0
5	1.0	0.0
6	0.0	1.0
7	0.0	1.0
8	0.0	0.0
9	0.0	0.0
10	0.6	0.2
11	0.0	0.0
12	0.0	0.0

A minimal cover found is given by the set $C = \{3, 5, 10\}$ with weights $(8, 11, 5)$ respectively. Adding the corresponding cover cut to the model and re-solving gives:

Table 2: Optimization Results (Objective Value: 194.2)

Index (i)	**Variable 1** ($j = 1$)	**Variable 2** ($j = 2$)
1	0.6	0.4
2	0.0	0.0
3	1.0	0.0
4	0.0	0.0
5	1.0	0.0
6	0.0	1.0
7	0.0	1.0
8	0.0	0.0
9	0.0	0.0
10	0.0	0.8
11	0.0	0.0
12	0.0	0.0

A second minimal cover found is given by the set $C = \{3, 5, 1\}$ with weights $(8, 11, 5)$ respectively. Adding the corresponding cover cut aswell to the model and re-solving gives:

Table 3: Optimization Results (Objective Value: 194.2)

Index (i)	**Variable 1** ($j = 1$)	**Variable 2** ($j = 2$)
1	1.0	0.0
2	0.0	0.0
3	0.0	1.0
4	0.0	0.0
5	1.0	0.0
6	0.0	1.0
7	0.6	0.4
8	0.0	0.0
9	0.0	0.0
10	0.0	0.8
11	0.0	0.0
12	0.0	0.0

Which seems to not improve the objective value... The cuts seems ineffective in this case.

Part 2

Introducing sequential lifting for this problem seems to not improve the objective value aswell. This is likely due to lifted cuts may not be strong enough, as we are only lifting one variable at a time.

Question 3

We know that:

$$\sum_{i=1}^{n} a_i x_i + \sum_{g_i<0} g_i y_i \leq \sum_{i=1}^{n} a_i x_i + \sum_{i=1}^{p} g_i y_i \leq b$$

$$\sum_{i=1}^{n} \left(\lfloor a_i \rfloor + f_i\right) x_i + \sum_{g_i<0} g_i y_i \leq \lfloor b \rfloor + f_0$$

$$\implies \sum_{i=1}^{n} f_i x_i + \sum_{g_i<0} g_i y_i - f_0 \leq \lfloor b \rfloor - \sum_{i=1}^{n} \lfloor a_i \rfloor x_i$$

Here we can clearly see that the RHS is an integer:

$$\sum_{i=1}^{n} f_i x_i + \sum_{g_i<0} g_i y_i - f_0 \leq K$$

Thus let it be denoted as K and lets see how it behaves for cases when $K \leq -1$ and $K \geq 0$.
Case $K \leq -1$:

$$\begin{aligned}
\sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i \leq \sum_{i=1}^{n} f_i x_i + \sum_{g_i<0} g_i y_i \leq K + f_0 \leq K(1 - f_0) \\
\implies \frac{1}{1-f_0}\left(\sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i\right) \leq K
\end{aligned}$$

This is because $(f_i - f_0)^+ \leq f_i$, and also when $K \leq -1$ then $K + f_0 \leq K(1 - f_0)$.
Case $K \geq 0$:

$$\begin{aligned}
\sum_{i=1}^{n} f_i x_i + \sum_{g_i<0} g_i y_i - f_0 \leq K \\
\implies \sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i \leq K + f_0
\end{aligned}$$

Now lets split up the LHS based on the set $S = \{i | f_i > f_0\}$ and start with the subcase where $\sum_{i \in S} x_i \geq K + 1$:

$$\begin{aligned}
\sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i = \sum_{i \in S} (f_i - f_0) x_i + \sum_{g_i<0} g_i y_i \leq \sum_{i \in S} f_i x_i + \sum_{g_i<0} g_i y_i \\
\implies \sum_{i \in S} f_i x_i + \sum_{g_i<0} g_i y_i \leq K + f_0 - f_0\left(\sum_{i \in S} x_i\right) = K + f_0\left(1 - \sum_{i \in S} x_i\right) \\
\implies \sum_{i \in S} f_i x_i + \sum_{g_i<0} g_i y_i \leq K + f_0\left(1 - \sum_{i \in S} x_i\right) \leq K + f_0(1 - (K+1)) = K(1 - f_0) \\
\implies \frac{1}{1-f_0}\left(\sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i\right) \leq K
\end{aligned}$$

For the other subcase where $\sum_{i \in S} x_i \leq K$, we have:

$$\begin{aligned}
\sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i = \underbrace{\sum_{i \in S} (f_i - f_0) x_i + \sum_{g_i<0} g_i y_i < \sum_{i \in S} (1 - f_0) x_i + \sum_{g_i<0} g_i y_i}_{f_i - f_0 < 1 - f_0} \\
\implies \frac{1}{1-f_0}\left(\sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i\right) < \sum_{i \in S} x_i + \underbrace{\frac{1}{1-f_0}\sum_{g_i<0} g_i y_i}_{\text{Negative}} \leq \sum_{i \in S} x_i \leq K \\
\implies \frac{1}{1-f_0}\left(\sum_{i=1}^{n} (f_i - f_0)^+ x_i + \sum_{g_i<0} g_i y_i\right) < K
\end{aligned}$$

Thus we can conclude that the LHS:

$$\frac{1}{1-f_0}\left(\sum_{i=1}^{n}(f_i-f_0)^+x_i+\sum_{g_i<0}g_iy_i\right)\leq K \qquad \forall K\in\mathbb{Z}$$

And thus substituting back for K gives us the final valid inequality:

$$\implies \frac{1}{1-f_0}\left(\sum_{i=1}^{n}(f_i-f_0)^+x_i+\sum_{g_i<0}g_iy_i\right)\leq\lfloor b\rfloor-\sum_{i=1}^{n}\lfloor a_i\rfloor x_i$$

$$\implies \sum_{i=1}^{n}\left(\lfloor a_i\rfloor+\frac{(f_i-f_0)^+}{1-f_0}x_i\right)+\sum_{g_i<0}\frac{g_i}{f_0-1}y_i\leq\lfloor b\rfloor$$

Question 4

Part 1

We have:

$$\sum_{i=1}^{n}a_ix_i+\sum_{i=1}^{p}g_iy_i=b$$

$$\sum_{i=1}^{n}(\lfloor a_i\rfloor+f_i)x_i+\sum_{i=1}^{p}g_iy_i=\lfloor b\rfloor+f_0$$

$$\implies \sum_{i=1}^{n}f_ix_i+\sum_{i=1}^{p}g_iy_i=f_0+\lfloor b\rfloor-\sum_{i=1}^{n}\lfloor a_i\rfloor x_i$$

$$\implies \sum_{f_i\leq f_0}f_ix_i+\sum_{f_i>f_0}(f_i-1)x_i+\sum_{i=1}^{p}g_iy_i=f_0+\underbrace{\lfloor b\rfloor-\sum_{i=1}^{n}\lfloor a_i\rfloor x_i-\sum_{f_i>f_0}x_i}_{K}$$

$$\implies \sum_{f_i\leq f_0}f_ix_i-\sum_{f_i>f_0}(1-f_i)x_i+\sum_{i=1}^{p}g_iy_i=f_0+K$$

Thus we once again get cases where $K\leq-1$ and $K\geq0$.
Case $K\geq0$:

$$\implies \sum_{f_i\leq f_0}\frac{f_i}{f_0}x_i-\sum_{f_i>f_0}\frac{(1-f_i)}{f_0}x_i+\sum_{i=1}^{p}\frac{g_i}{f_0}y_i=1+\frac{K}{f_0}$$

$$\implies \sum_{f_i\leq f_0}\frac{f_i}{f_0}x_i-\sum_{f_i>f_0}\frac{(1-f_i)}{f_0}x_i+\sum_{g_i<0}\frac{g_i}{f_0}y_i+\sum_{g_i>0}\frac{g_i}{f_0}y_i\geq1$$

Case $K\leq-1$:

$$\implies \sum_{f_i\leq f_0}f_ix_i-\sum_{f_i>f_0}(1-f_i)x_i+\sum_{i=1}^{p}g_iy_i=f_0+K\leq K(1-f_0)$$

$$\implies \sum_{f_i\leq f_0}\frac{f_i}{1-f_0}x_i-\sum_{f_i>f_0}\frac{(1-f_i)}{1-f_0}x_i+\sum_{i=1}^{p}\frac{g_i}{1-f_0}y_i\leq K\leq-1$$

$$\implies -\sum_{f_i\leq f_0}\frac{f_i}{1-f_0}x_i+\sum_{f_i>f_0}\frac{(1-f_i)}{1-f_0}x_i-\sum_{g_i<0}\frac{g_i}{1-f_0}y_i-\sum_{g_i>0}\frac{g_i}{1-f_0}y_i\geq1$$

We can thus see that in both cases we have a linear inequality of the form:

$$\alpha z \geq 1$$

Which implies that $\max(\alpha_1^i, \alpha_2^i) \cdot z^i \geq 1$, where α_1^i and α_2^i are the coefficients from the two cases above respectively and the superscript i correspond to a coefficient in the system. Thus we can combine them to get the final valid inequality:

$$\sum_{f_i \leq f_0} \frac{f_i}{f_0} x_i + \sum_{f_i > f_0} \frac{(1-f_i)}{1-f_0} x_i - \sum_{g_i < 0} \frac{g_i}{1-f_0} y_i + \sum_{g_i > 0} \frac{g_i}{f_0} y_i \geq 1$$
$$\sum_{f_i \leq f_0} \frac{f_i}{f_0} x_i + \sum_{f_i > f_0} \frac{1-f_i}{1-f_0} x_i + \sum_{g_i > 0} \frac{g_i}{f_0} y_i - \sum_{g_i < 0} \frac{g_i}{1-f_0} y_i \geq 1$$

Part 2

Based on Section 4.9, intersection cuts provide a geometric approach to generating valid inequalities for mixed integer linear programs. The process begins by analyzing the corner polyhedron, which is a simplified view of the problem's feasible region. To create this, we take the optimal solution to the linear relaxation, denoted as $\bar{x}$, and ignore the non-negativity constraints on the basic variables. This leaves us with a cone-like shape where the fractional solution $\bar{x}$ sits at the vertex, formed by the constraints that are active at that point.

Because $\bar{x}$ is fractional (not an integer), it falls into a "forbidden" gap known as a split disjunction. This is a region defined by two parallel hyperplanes, such as $\pi x \leq \pi_0$ and $\pi x \geq \pi_0 + 1$. The space between these planes contains $\bar{x}$ but cannot contain any valid integer solutions.

To generate the cut, we examine the rays (directions denoted as r^j) that extend outward from the vertex $\bar{x}$ along the edges of the cone. For each ray, we calculate a distance scalar α_j. This value α_j represents exactly how far one must travel along that specific ray to leave the forbidden region and intersect with one of the valid boundaries.

The final intersection cut is expressed as the inequality:

$$\sum_{j \in J} \frac{x_j}{\alpha_j} \geq 1$$

This formula dictates that to reach a valid integer solution, the non-basic variables x_j must increase enough so that their weighted sum moves the point out of the forbidden gap. This effectively slices off the corner of the cone containing $\bar{x}$ while ensuring all feasible integer points remain valid.

Question 5

Check the code. But we can see that adding the GMI cuts increases the objective value from 265.7 to 273.3 when the true optimal integer solution would yield 285.